

The Body Bus

A wearable computer will be useless if you have to walk around looking like the back of your desk. Fortunately, bits are more than skin deep. Tom Zimmerman has shown that the non-contact coupling between your body and weak electric fields can be used to create and sense tiny nano-amp currents in your body. Modulating these signals creates Body Net, a personal-area network that communicates through your skin. Using roughly the same voltage and frequencies as audio transmissions, this will be as safe as wearing a pair of headphones. Keeping data in your body avoids the intrusion of wires, the need for an optical path for infrared, and conventional problems such as regulation and eavesdropping.

Your shoe computer can talk to a wrist display and keyboard and heads up glasses. Activating your body means that everything you touch is potentially digital. A handshake becomes an exchange of digital business cards, a friendly arm on the shoulder provides helpful data, and picking up a phone downloads your numbers and voice signature for faithful speech recognition.

Cyborgs

Cyborgs are here already. No, this isn't a paranoid fantasy about intruders from the future. Two cyborgs have been roaming the Media Lab, wearing computers day in and day out for over two years. It's an uncanny experience teaching a course to Thad Starner, who is simultaneously watching you lecture and annotating the lecture notes behind you through Reflection Technologies' Private Eye, a wearable heads-up display.

Steve Mann goes further, wearing a completely immersive system: movable cameras connect to a computer and transmitter to send video to a workstation for processing and delivery back to displays in front of his eyes.

This lets him enhance what he sees (he likes living in a 'rot 90' rotated world) and position his eyes. (Some days he likes having his eyes above his head, or at his feet, and when he rides a bicycle he sets one eye looking forward and one backward.)

He can assemble everything he's seen into larger mosaics or 3-D images, and through the radio-frequency link you can see through his eyes (at www.wearcam.org).

Don't expect to see much computing featured in Bill Blass' next collection, but this kind of digital headdress will become more common. Bear in mind that 20 years ago, no publisher anticipated that teletype terminals would grow into a portable threat to books, that paper tapes would merge with film into multimedia CD-ROMs, or that telephones would threaten the whole business model of publishing by bringing the Web into your home.

The difference in time between loony ideas and shipped products is shrinking so fast that it's now, oh, about a week.

This article was co-authored by Neil Gershenfeld, an MIT professor and one of three co-principal investigators of the Media Lab's newest research consortium, Things That Think.

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